

Noah Martino

Software Engineer | Backend, Product, and Full-Stack Engineering

Pittsburgh, PA · 412-304-6414 · personal.boredhero@gmail.com

linkedin.com/in/noah-martino · github.com/boredhero · noahmartino.com

SUMMARY

Founding engineer who helped build a financial kiosk product from prototype to thousands of active devices nationwide. Builds and owns backend services, full-stack applications, internal tools and support portals, and Linux deployment and operations tooling.

EXPERIENCE

Freedom Gateway — Pittsburgh, PA

Lead Software Engineer | May 2024–Present

Software Engineer | Independent Contractor, Part-time | Oct 2019–Jul 2021; May 2023–Apr 2024

- Built the central Python/PostgreSQL kiosk backend and core business logic; provide technical direction for a small team of engineers.
- Built kiosk hardware-integration and monitoring services with modern Python runtimes and native dependencies packaged for legacy Linux, including peripheral APIs, signed MQTT commands, health telemetry, and automated service recovery.
- Built a FastAPI artifact-distribution backend delivering software updates to thousands of IoT devices nationwide, with expiring, usage-limited download links, audit logging, and atomic replacement to prevent serving partially uploaded files.
- Built a mobile-first fleet operations and support platform with Flask and React, combining live fleet monitoring, browser-based SSH/VNC, remote configuration, installation/update diagnostics, product analytics, and artifact sharing.
- Built a full-stack operations portal with Angular, FastAPI, and PostgreSQL, including organization-scoped access, automated reporting, PDF statements, and cross-worker caching.
- Built a custom interactive Linux installer image for warehouse imaging and field recovery; automated encrypted OS installation, hardware-specific service setup, data backup and restoration, and bootable-image releases.
- Built a remote-access platform with a Python/PostgreSQL WireGuard control plane and Rust CLI, including peer recovery and connection fallback, avoiding higher vendor subscription costs.
- Built and administered a multi-node on-prem Proxmox cluster behind OPNsense; oversaw dedicated fiber installation.
- Built CI/CD pipelines across full-stack and backend projects with self-hosted runners to reduce costs; custom Docker images and layered caching cut full runs on multiple pipelines from ~20 minutes to under 3 minutes with warm caches.
- Created an agent-assisted kiosk troubleshooting workflow that gathers logs and traces issues across source code and services; teammates use it in place of most manual kiosk-log review.
- Trained and deployed a YOLO model on 5,000+ human-reviewed kiosk images, enabling automatic camera-role configuration with inference under 30 ms on resource-constrained kiosk hardware.
- Designed a unified kiosk configuration system with typed settings and automatic migration across software releases.
- Implemented the kiosk application's software-update interface and backend, with download progress, stall detection, failure recovery, and installation scheduled around customer transactions.
- Implemented Wi-Fi geolocation across kiosk clients, frontend maps, and backend services to track deployed devices, minimizing Google API costs with a 97.4% cache hit rate.
- Implemented automatic legacy JSON-to-SQLite database conversion during kiosk upgrades, with recovery safeguards.

SELECTED PERSONAL PROJECTS

Plant Camera System · Python, Flask, React, HLS.js, FFmpeg, Docker, Linux

Designed a two-stage video filter that removes 120 Hz LED/rolling-shutter banding without ghosting, using approximately 64% less CPU than the original 31-frame filter. Built live HLS video, timelapse, device-identity, and recovery workflows.

More Fuels Mod · Java, Minecraft Forge

Built and maintained an open-source mod across 13 Minecraft versions, reaching more than 272,000 verified CurseForge downloads.

TECHNICAL SKILLS

Languages: Python, Java, Bash, SQL, Rust, TypeScript

Application & Data: FastAPI, Flask, JavaFX, React, Angular, PostgreSQL, Redis, Memcached, SQLite, Firestore, SQLAlchemy, WebSockets, OpenCV, ONNX

Platform & Delivery: Linux, Docker/Swarm, GitHub Actions, Git, Cython, WireGuard, MQTT, systemd, Nginx, Apache HTTP Server, Proxmox, OPNsense, OpenTelemetry

Engineering: Distributed systems, API design, performance optimization, observability, automated testing, hardware integration, AI-assisted development

EDUCATION

University of Pittsburgh — Bachelor of Science, Digital Narrative and Interactive Design, Apr 2024

Interdisciplinary computing program in software development, information science, human-computer interaction, and design.